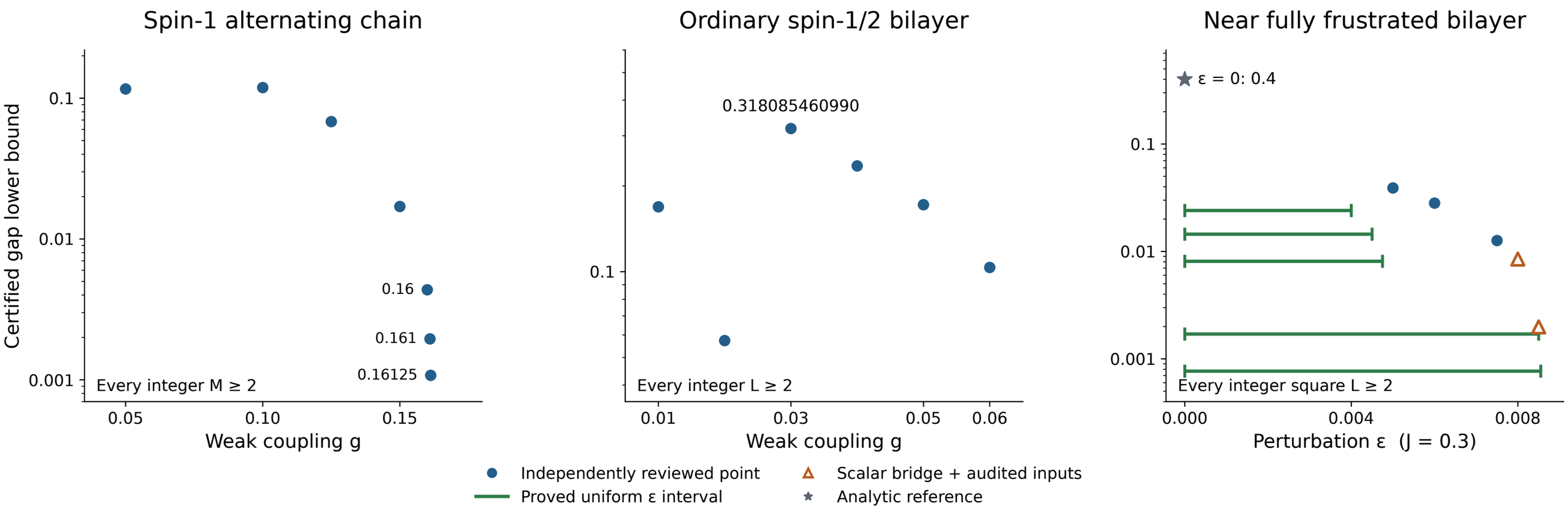
Spin-1 alternating chain
Ordinary spin-1/2 bilayer
Near fully frustrated bilayer
Certified gap lower bound
0.1
0.01
0.001
0.16
0.161
0.16125
Every integer M ≥ 2
0.05
0.10
0.15
Weak coupling g
0.318085460990
0.1
Every integer L ≥ 2
0.01
0.03
0.05
0.06
Weak coupling g
ε = 0: 0.4
0.1
0.01
0.001
Every integer square L ≥ 2
0.000
0.004
0.008
Perturbation ε (J = 0.3)
Independently reviewed point
Proved uniform ε interval
Scalar bridge + audited inputs
Analytic reference
Points certify individual parameters; horizontal bars certify entire intervals.
Different certificates can yield nonmonotone bounds across g; no physical-gap trend is inferred.